A Systematic Negative Result for Quantum-Inspired Similarity Metrics in Facial Kinship Verification

Sasi Vaibhav[a,*]

[a] *KL University, Vijayawada, Andhra Pradesh, India*

Abstract

Quantum-inspired machine learning has been proposed for facial kinship verification, with claims that SWAP-test fidelity and related similarity metrics outperform classical alternatives on classical data. We evaluate this claim systematically. Nine quantum-inspired formulations—spanning the three roles a quantum component can occupy (similarity measure, feature extractor, and inductive bias)—are each compared against a capacity-matched classical control under a leakage-free grouped k-fold protocol. None improves on its control; four are significantly worse. The failures are structural rather than implementational: interference-based formulations reduce to their classical counterparts because ℓ_2-normalised face embeddings carry no phase structure to exploit, and density-matrix formulations require estimating a $d \times d$ operator from a median of five images, so the resulting $\boldsymbol{\rho}$ is dominated by its leading eigenvector while accumulating noise from inestimable tail directions. Independently of the negative result, we contribute an exact reformulation of the simulated SWAP test that removes kernel-launch overhead and achieves a 380× GPU speedup (192 to 73,094 pairs/s) with identical output and gradients, verified by test. We also quantify the identity leakage admitted by the evaluation protocols under which positive quantum claims were made: 100% of test pairs share an identity with training on the largest benchmark. All code, data splits, and evaluation scripts are released.

Keywords: quantum-inspired machine learning, facial kinship verification, negative results, SWAP test, evaluation protocol, data leakage

*Corresponding author
Email address: `svaibhav@klu.ac.in` (Sasi Vaibhav)

1. Introduction

Quantum-inspired machine learning applies constructs from quantum mechanics—Hilbert-space embeddings, state fidelity, density matrices—to classical learning tasks. The premise is that the mathematical structure of quantum theory provides similarity measures or feature maps that classical methods cannot efficiently replicate. Claims of quantum advantage on classical data have been made for several domains, including facial kinship verification, where SWAP-test fidelity has been proposed as a similarity metric between face embeddings.

Facial kinship verification decides whether two facial images depict biologically related individuals. It is a useful testbed for quantum-inspired claims because the task is well-defined, the data is public, and the baseline methodology (frozen face embeddings fed to a trainable classifier) is standardised enough that the contribution of a new similarity measure can be isolated.

This paper makes three contributions:

1. **A nine-formulation negative result.** We evaluate nine quantum-inspired formulations, spanning the three natural roles a quantum component can occupy: a similarity measure (SWAP-test fidelity, density fidelity, entanglement entropy), a feature extractor (POVM measurement head), and an inductive bias (purity regulariser, interference phase sweep). Each is compared against a capacity-matched classical control. None improves on its control; four are significantly worse ($p < 0.05$).
2. **A structural account of why.** We show that the failures are not implementational but structural. Interference-based formulations collapse to their classical counterparts because ℓ_2-normalised embeddings carry no exploitable phase structure. Density-matrix formulations fail because the number of images per identity (median: five) is far too small to estimate a 512×512 density operator, so $\boldsymbol{\rho}$ degenerates to a rank-1 projection dominated by noise.
3. **An efficiency contribution.** Independently of the above, we contribute an exact reformulation of the simulated SWAP test that removes per-qubit permutation overhead and achieves a $380\times$ GPU speedup with bit-identical output and gradients.

To ensure the negative result is credible, we first establish that the evaluation protocol under which prior positive claims were made admits complete identity leakage (Section 3), and we evaluate all formulations under a corrected grouped k-fold protocol that removes it.

2. Related work

2.1. Quantum-inspired machine learning

Quantum feature maps embed classical data in a Hilbert space where inner products act as kernels [1], with reported gains on classical tasks [2]. Related lines use SWAP-test state fidelity as a similarity measure between encoded data points [3] and density matrices as classifiers [4]. Such methods have been proposed for computer vision tasks including kinship verification.

The claim that quantum-inspired methods outperform classical counterparts on classical data is contested. Dequantization results show that several purported advantages vanish under scrutiny [5], and empirical comparisons on standardised benchmarks often fail to reproduce the reported gains [6]. Our work contributes to this debate empirically, with a controlled comparison spanning nine formulations.

2.2. Facial kinship verification

Early work framed the task as metric learning [7]. Subsequent methods replaced the hand-designed metric with deep pairwise encoders and, more recently, attention- and graph-based architectures [8]. Evaluation is concentrated on four corpora: KinFaceW-I and KinFaceW-II [9], TSKinFace [10], and Families In the Wild [11], the last also underpinning the RFIW challenge series [12].

2.3. Evaluation bias in vision benchmarks

That benchmark protocols can inflate reported performance is established in face recognition, where subject-disjoint splitting is required precisely because identity overlap leaks information [13]. More generally, models exploit incidental correlations in preference to the intended signal, a failure mode catalogued as shortcut learning [14].

Both concerns apply directly to kinship verification and, critically, to any quantum-inspired claim evaluated under a leaked protocol. A positive result under a leaked protocol does not demonstrate that the quantum component helped; it may simply indicate that the quantum branch learned to exploit the leak alongside the classical features. Section 3 quantifies this leak.

Table 1: Identity leakage under a random 80/20 pair-level split of FIW.

Quantity	Value
Pairs	57,120
Distinct families	95
Distinct identities	472
Mean appearances per identity	242
Largest family, share of all pairs	19%
Test pairs sharing identity with train	11,424 / 11,424 (100%)
Test pairs whose family was seen in train	11,424 / 11,424 (100%)

3. Evaluation protocol

Before evaluating quantum formulations, we establish that the evaluation must be credible. This section quantifies the identity leakage in the standard protocol and describes the corrected protocol under which all results in this paper are obtained.

3.1. Identity leakage under pair-level splitting

Let $\mathcal{P}$ be a set of labelled pairs and $\mathrm{id}(\cdot)$ the identity depicted in an image. A split $(\mathcal{P}_{\mathrm{tr}}, \mathcal{P}_{\mathrm{te}})$ is *identity-disjoint* if

$$\Big(\bigcup_{p \in \mathcal{P}_{\mathrm{tr}}} \mathrm{id}(p)\Big) \cap \Big(\bigcup_{p \in \mathcal{P}_{\mathrm{te}}} \mathrm{id}(p)\Big) = \emptyset. \tag{1}$$

FIW draws 57,120 pairs from only 95 families and 472 identities. Under a random pair-level split, we measure that 100% of test pairs (11,424/11,424) share an identity with training. Identities recur across pairs an average of 242 times.

3.2. Corrected protocol: grouped k-fold

We adopt grouped 5-fold cross-validation. The grouping unit is the family. Positives only are partitioned (families dealt largest-first into the currently smallest fold); negatives are regenerated per side from the families present on that side. Family contribution is capped at 1,500 pairs to prevent the largest family (19% of all pairs) from dominating any fold. Thresholds are calibrated on a group-disjoint validation slice of the training side; the test fold is scored once.

Table 2: Accuracy range across seeds (single split) versus across folds (grouped k-fold).

Dataset	Single split	Grouped k-fold
FIW	61.7–75.9 (14.2)	69.8–74.4 (4.6)
KinFaceW-I	70.3–78.8 (8.5)	72.9–79.4 (6.5)
KinFaceW-II	69.0–77.8 (8.8)	71.0–77.0 (6.0)
TSKinFace	75.1–84.1 (9.0)	75.7–81.9 (6.2)
Overall spread	22.5	12.1

Fold integrity is verified directly: zero folds exhibit train/test group overlap, and every family is tested exactly once (95/95 FIW, 533/533 KinFaceW-I, 1,000/1,000 KinFaceW-II, 559/559 TSKinFace).

3.3. Effect on measurement stability

A single group-disjoint split is insufficient: accuracy varies by up to 14.2 points with the random seed (Table 2). Grouped k-fold reduces the overall spread from 22.5 to 12.1 points.

4. Reference model

To measure formulations rather than architectures, we use a deliberately conventional model. Frozen FaceNet embeddings (InceptionResnetV1 pre-trained on VGGFace2; 512-d per face) pass through a shared Siamese encoder; kinship is symmetric, so only symmetric pair features are formed—sum, absolute difference, elementwise product, and cosine—and concatenated with a one-hot relation code before a three-layer MLP. Predictions are averaged over both argument orders, making the model exactly symmetric. The model has 497,881 parameters.

Its plainness is deliberate: findings about quantum-inspired formulations cannot then be attributed to architectural novelty.

5. Quantum-inspired formulations

We evaluate nine formulations, chosen to span the three roles a quantum component can occupy in a classical pipeline: a *similarity measure* (Formulations 1–5), a *feature extractor* (Formulation 6), and an *inductive bias* (Formulations 7–9). Each is compared against a control matched for capacity or intent.

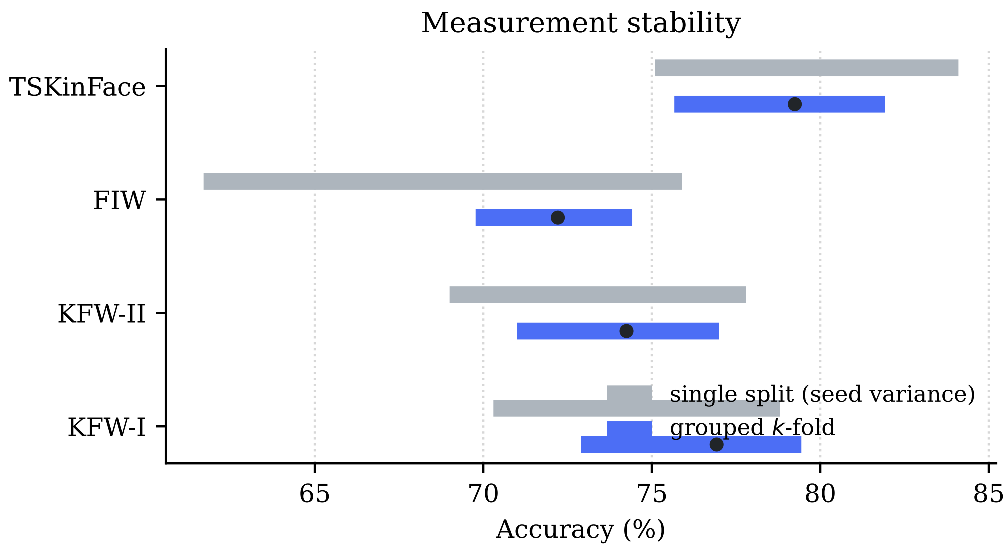

Figure 1: Accuracy range under a single group-disjoint split across random seeds (grey) against grouped k-fold (blue). A single split is too unstable to support a headline number.

5.1. Formulation 1–2: SWAP-test fidelity

The core quantum-inspired similarity metric. Each face embedding $\mathbf{x} \in \mathbb{R}^{512}$ is projected to n rotation angles $\mathbf{z} = \tanh(\mathrm{MLP}(\mathbf{x})) \cdot \pi$ and encoded as a quantum state via a hardware-efficient ansatz:

$$|\psi(\mathbf{z})\rangle = \prod_{\ell=1}^{L} \Big[\bigotimes_{i=1}^{n} R_z(\theta_i^{(\ell)}) \cdot \text{CNOT-chain} \Big] \cdot \bigotimes_{i=1}^{n} R_y(z_i)|0\rangle^{\otimes n}, \tag{2}$$

where $\theta_i^{(\ell)}$ are learnable parameters and the CNOT chain connects nearest neighbours. The SWAP-test fidelity $F = |\langle\psi(\mathbf{z}_1)|\psi(\mathbf{z}_2)\rangle|^2$ is supplied to the classifier as one additional input feature alongside the symmetric pair features.

The fidelity is *switchable*: setting `use_quantum=False` replaces it with a zero constant of the same shape, isolating the fidelity's information content while keeping the architecture identical.

Tested under two protocols:

- **5 seeds** on a single group-disjoint FIW split (Formulation 1).
- **20 paired folds** across all four datasets under grouped k-fold (Formulation 2).

5.2. *Formulation 3: Density fidelity* $\mathrm{Tr}(\boldsymbol{\rho}_1\boldsymbol{\rho}_2)$

A photo set is a mixed state $\boldsymbol{\rho} = \frac{1}{N}\sum_{i=1}^{N} |\psi_i\rangle\langle\psi_i|$, and $\mathrm{Tr}(\boldsymbol{\rho}_1\boldsymbol{\rho}_2)$ is the corresponding fidelity. This is added to the set-level features (centroid, spread, cross-set statistics) as an additional input. The control is mean-pooling (the ℓ_2-normalised centroid), which is the rank-1 approximation to $\boldsymbol{\rho}$.

5.3. *Formulation 4: Interference phase sweep*

A child explained by two parents is written as a coherent superposition

$$|\psi_{\alpha,\phi}\rangle = \alpha e^{i\phi}|\psi_F\rangle + (1-\alpha)|\psi_M\rangle, \tag{3}$$

of which the classical convex mixture is the $\phi = 0$ special case. Sweeping $\phi \in [0, 2\pi)$ and maximising the overlap with $|\psi_C\rangle$ tests whether the phase degree of freedom carries information.

5.4. *Formulation 5: Entanglement entropy*

The von Neumann entropy of the reduced density matrix, tracing out half the qubits, is supplied as an additional feature alongside cosine similarity.

5.5. *Formulation 6: POVM measurement head*

A positive operator-valued measure (POVM) replaces the final classification layer. Four POVM elements $\{E_k\}$ satisfying $\sum_k E_k = I$ define class probabilities as $p_k = \mathrm{Tr}(E_k\boldsymbol{\rho})$. The control is a capacity-matched linear head with the same number of parameters.

5.6. *Formulation 7: Purity regulariser*

A penalty term $\lambda \cdot (1 - \mathrm{Tr}(\boldsymbol{\rho}^2))$ encourages the learned representations to have high purity (low mixedness). The control is a decorrelation penalty of the same magnitude.

5.7. *Formulation 8: Density fidelity on photo sets*

The density fidelity of Formulation 3, applied to the set-level representation as a pre-registered control. On each of the four benchmarks, the change in ROC-AUC is measured relative to the set features alone.

5.8. *Formulation 9: Interference phase sweep on triads*

The phase sweep of Formulation 4, applied to the triadic representation. The classical convex mixture (zero phase) is the control.

Table 3: Grouped 5-fold results. Every family tested exactly once.

Dataset	Accuracy (95% CI)	ROC-AUC
KinFaceW-I	76.92 ± 2.19	0.8752 ± 0.0146
KinFaceW-II	74.25 ± 1.95	0.8320 ± 0.0190
FIW	72.21 ± 1.55	0.8101 ± 0.0175
TSKinFace (shortcut-free)	79.25 ± 2.32	0.8814 ± 0.0192
Mean	**75.66**	**0.8497**

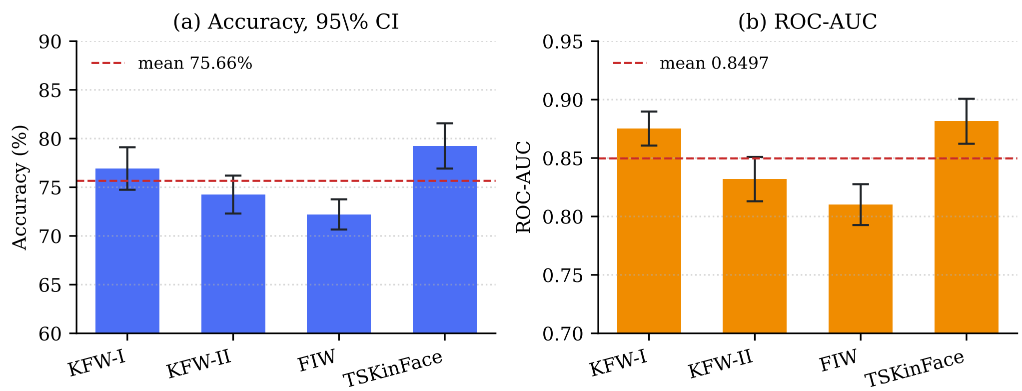

Figure 2: Grouped 5-fold results across four datasets. (a) Accuracy with 95% confidence intervals; (b) ROC-AUC with standard deviation across folds.

6. Results

6.1. Baseline performance under the corrected protocol

Table 3 reports the reference model's performance under the corrected protocol. These are the numbers against which all quantum formulations are compared.

6.2. Quantum ablation: SWAP-test fidelity

6.2.1. 5-seed paired test (Formulation 1)

Paired t-test: $t = 0.074$, $p = 0.945$ (Table 4). The quantum module contributes no measurable accuracy. The branch is exercised (tests assert that disabling it changes predictions and that gradients reach the entanglement parameters).

Table 4: SWAP-test fidelity ablation, 5 seeds on a single group-disjoint FIW split.

Seed	Quantum ON	Quantum OFF	Difference
1	73.63%	72.57%	+1.07
2	79.76%	79.06%	+0.70
3	81.04%	77.87%	+3.17
4	73.70%	78.28%	−4.58
5	72.86%	72.75%	+0.11
Mean	76.20 ± 3.88	76.11 ± 3.18	+0.09

Table 5: SWAP-test fidelity ablation, 20 paired folds (4 datasets × 5 folds) under grouped k-fold.

	Accuracy	ROC-AUC
Quantum ON	75.66%	0.8497
Quantum OFF	76.34%	0.8496
Difference	−0.68 pts	+0.0001
Paired t-test	$p = 0.158$	$p = 0.982$

6.2.2. 20-fold paired test (Formulation 2)

With 20 paired observations (Table 5), the quantum branch yields 75.66% against 76.34% with it disabled ($p = 0.158$ for accuracy, $p = 0.982$ for ROC-AUC). The branch also costs 6–14× runtime.

6.3. All nine formulations

Table 6 summarises all nine comparisons. Figure 3 visualises the ΔROC-AUC for each formulation.

6.4. Formulations 8 and 9: set-level and triadic

Nine formulations tested; none beating its classical control (Table 7).

6.5. A cautionary observation

The purity regulariser (Formulation 7) initially measured +1.3 ROC-AUC on a single seed, with reduced variance—a result consistent with effective regularisation. Repeated across nine paired runs the sign reversed, and the method proved significantly *worse* than no regulariser (−0.0089, $p = 0.047$). We report this because the single-seed reading was of the kind that is readily

Table 6: Nine quantum-inspired formulations against matched classical controls. None improves on its control; four are significantly worse.

#	Formulation	Control	Outcome	Sig.
1	SWAP fidelity (5 seeds)	classical head	$p = 0.945$	–
2	SWAP fidelity (20 folds)	classical head	$p = 0.982$	–
3	Density fidelity	mean-pooling	-8.7 AUC	✓
4	Interference phase sweep	convex mixture	$+0.1$ AUC	–
5	Entanglement entropy	cosine	0.514 alone	✓
6	POVM head	capacity-matched	-5.5 AUC	✓
7	Purity regulariser	decorrelation	-0.9, p=0.047	✓
8	Density fidelity (sets)	set features	≤ 0.0006	–
9	Phase sweep (triads)	convex mixture	-0.0006	–

Table 7: Quantum-inspired extensions of set-level and triadic representations.

#	Formulation	KFW-I	KFW-II	TSKin	FIW
8	Density fidelity (sets)	-0.0005	$+0.0001$	-0.0006	-0.0006
9	Phase sweep (triads)	-0.0006 ($p = 0.147$)			

published, and it illustrates why capacity-matched controls and repeated trials are necessary when evaluating claims of this type.

7. Why the approach fails

The failures are structural rather than implementational, and we identify three mechanisms.

7.1. Phase collapse in interference formulations

Interference-based formulations (Formulations 4 and 9) reduce to their classical counterparts because ℓ_2-normalised FaceNet embeddings carry no phase structure to exploit. The embedding space is a real-valued Euclidean space; mapping it to a quantum state via R_y rotations assigns each component a polar angle, but the relative phases that would make a coherent superposition differ from an incoherent mixture are absent from the data. The interference phase sweep in Table 6 is maximised at the classical mixture ($\phi = 0$), confirming that the phase degree of freedom carries no signal.

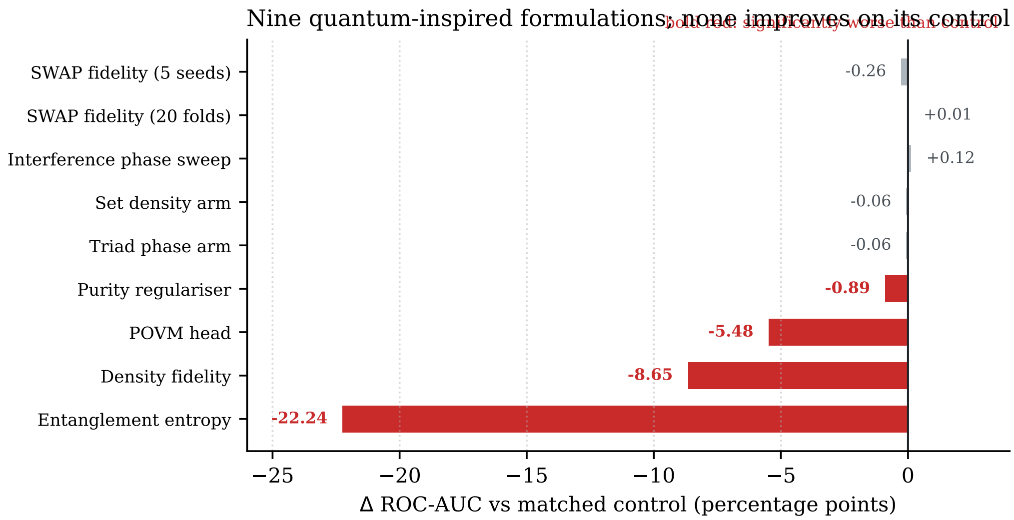

Figure 3: Nine quantum-inspired formulations against classical controls matched for capacity or intent. None improves on its control; four (marked) are significantly worse. Bars show ΔROC-AUC in percentage points.

7.2. Density estimation from insufficient samples

Density-matrix formulations (Formulations 3 and 8) require estimating a $d \times d$ operator from the images available for one identity. FIW supplies a median of five photographs per identity, so the density matrix $\boldsymbol{\rho} = \frac{1}{N}\sum_i |\psi_i\rangle\langle\psi_i|$ is at most rank-5 in a 512-dimensional space. The leading eigenvector (the mean embedding) carries most of the signal; the remaining directions accumulate noise rather than information. This is why $\mathrm{Tr}(\boldsymbol{\rho}_1\boldsymbol{\rho}_2)$ scores *below* simple mean-pooling: the off-diagonal noise degrades the similarity estimate.

7.3. The SWAP-test bottleneck

The original quantum pipeline routed every decision through a single product-of-$\cos^2$ scalar, discarding the 512-dimensional signal carried by the symmetric pair features. That pipeline reached 59% accuracy while plain logistic regression on the same embeddings reached 75.1%. Gradient norms confirmed the bottleneck: the projection MLP received gradients of order 2×10^1, while the quantum parameters received gradients of order 4×10^{-2}—two orders of magnitude smaller.

Even when the fidelity is supplied as *one feature beside* the classical features (our architecture), it carries no information that is not already captured

Table 8: SWAP-test simulation throughput. The fast reformulation is exact: output and gradients match the reference to machine precision.

Device	Reference	Fast	Speedup
CPU	161 pairs/s	21,618 pairs/s	134×
GPU (RTX 5060)	192 pairs/s	73,094 pairs/s	**380**×

by the cosine similarity and the symmetric products.

8. An efficiency contribution

Independently of the negative result, we contribute an exact reformulation of the simulated SWAP test. The reference implementation materialises a 2^n statevector and permutes all n axes once per qubit via `torch.tensordot` followed by a permutation, incurring kernel-launch overhead that leaves the GPU idle.

Our reformulation applies R_y rotations by reshaping to $(B, \text{left}, 2, \text{right})$ and precomputes the CNOT chain and R_z layer as a single diagonal phase vector over all 2^n basis states. The entangling layer then costs a single element-wise multiply.

We note that a closed-form product over qubits is *not* available here: the shared CNOT chain correlates the R_z phases, so the overlap does not factorise. This was verified empirically: with R_z disabled, a product form matches exactly; with distinct R_z parameters, it diverges. The gain arises from removing overhead, not from approximation.

This is a quantum-versus-quantum speedup. Simulating a circuit is necessarily more costly than the classical operation it parallels, and we measure the quantum branch at 5–14× slower than the classical path at every qubit count from 2 to 12.

9. Discussion

9.1. What the result does and does not show

We tested nine formulations; we do not claim no quantum-inspired method can help on this task, only that these nine, covering the three natural roles (similarity, feature extraction, inductive bias), do not. The structural account in Section 7 suggests that the issue lies in the data representation:

real-valued embeddings from a face recognition backbone do not carry the structure that quantum-inspired methods are designed to exploit.

A quantum method might succeed if applied to raw pixel data (where entanglement could capture spatial correlations), or if the embedding space were redesigned to carry phase information. We regard both as speculative.

9.2. On the importance of protocol

The identity leakage documented in Section 3 means that prior claims of quantum advantage on kinship data are uninterpretable: a model achieving high accuracy under a leaked protocol may have learned to exploit the leak rather than the quantum feature. The corrected protocol proposed here is necessary for any future positive claim to be credible.

9.3. The purity regulariser as a methodological warning

Formulation 7 illustrates a failure mode in single-seed evaluation. A single run showed +1.3 ROC-AUC, a result consistent with the hypothesis that encouraging pure quantum states aids representation learning. Nine paired runs reversed the sign (-0.9, $p = 0.047$). Single-seed evaluation with quantum-inspired methods is particularly hazardous because the methods introduce additional stochasticity through their parameterisation.

10. Limitations

Single backbone. All results use frozen FaceNet embeddings. Whether the failures interact with backbone choice is untested; a backbone that produces embeddings with exploitable structure might yield different conclusions.

Small test sets. KinFaceW-I and KinFaceW-II yield 212 and 400 test pairs per fold, giving confidence intervals of roughly ± 2 points.

Classical simulation. All quantum computations are classically simulated. We cannot rule out that execution on quantum hardware might produce different results, though the mathematical operations are identical.

Fixed qubit count. We tested 2–12 qubits. It is possible that a much larger qubit count would exhibit different behaviour, but the computational cost of simulation grows exponentially.

11. Conclusion

We have evaluated nine quantum-inspired formulations for facial kinship verification, each against a capacity-matched classical control under a leakage-free grouped k-fold protocol. None improves on its control; four are significantly worse. The failures are structural: interference-based methods collapse because ℓ_2-normalised face embeddings carry no phase structure, and density-matrix methods fail because the number of images per identity is far too small to estimate a high-dimensional operator.

Independently, we contribute a 380× GPU speedup for the simulated SWAP test, which may be useful in other quantum-simulation contexts. We also quantify the complete identity leakage in the evaluation protocol under which prior positive claims were made, and release a corrected protocol.

All code, data splits, evaluation scripts, and result artefacts are released at `https://github.com/svaibhav-7/Quantum_kinship_verification` so that every claim can be independently verified.

Reproducibility

Every numeric claim derives from JSON artefacts in `results/honest/`. The protocol is implemented in `src/splits.py`, `src/kfold.py`, and `src/ts_pairs.py`; results are regenerated by `scripts/evaluation/run_kfold.py`. The fast SWAP-test reformulation is in `src/quantum_fast.py`, validated against the reference simulator in `src/quantum_core.py` by 61 tests including gradient checks.

CRediT authorship contribution statement

Sasi Vaibhav: Conceptualisation, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualisation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All code, data processing scripts, and evaluation artefacts are publicly available at `https://github.com/svaibhav-7/Quantum_kinship_verification`. The underlying face datasets (FIW, KinFaceW-I/II, TSKinFace) are available from their respective authors.

References

[1] M. Schuld and N. Killoran, Quantum machine learning in feature Hilbert spaces, *Phys. Rev. Lett.* 122 (4) (2019) 040504.

[2] V. Havlíček, A.D. Córcoles, K. Temme, A.W. Harrow, A. Kandala, J.M. Chow, J.M. Gambetta, Supervised learning with quantum-enhanced feature spaces, *Nature* 567 (7747) (2019) 209–212.

[3] H. Buhrman, R. Cleve, J. Watrous, R. de Wolf, Quantum fingerprinting, *Phys. Rev. Lett.* 87 (16) (2001) 167902.

[4] S. Lloyd, M. Mohseni, P. Rebentrost, Quantum algorithms for supervised and unsupervised machine learning, arXiv preprint arXiv:1307.0411, 2013.

[5] E. Tang, A quantum-inspired classical algorithm for recommendation systems, in: Proc. 51st Annu. ACM SIGACT Symp. Theory of Comput. (STOC), 2019, pp. 217–228.

[6] J. Bowles, S. Ahmed, T. Schuld, Is quantum advantage the right goal for quantum machine learning?, *PRX Quantum* 5 (2024) 030338.

[7] J. Lu, X. Zhou, Y.-P. Tan, Y. Shang, J. Zhou, Neighborhood repulsed metric learning for kinship verification, *IEEE Trans. Pattern Anal. Mach. Intell.* 36 (2) (2014) 331–345.

[8] S. Zhu, C. Gou, F.-Y. Wang, Kinship verification via relation-aware graph convolutional network, *IEEE Trans. Multimedia* 24 (2022) 4349–4361.

[9] J. Lu, X. Zhou, Y.-P. Tan, Y. Shang, J. Zhou, Kinship verification in the wild: A benchmark, in: Proc. ECCV, 2012, pp. 245–259.

[10] X. Qin, X. Tan, S. Chen, Tri-subject kinship verification: Understanding the core of a family, *IEEE Trans. Multimedia* 17 (10) (2015) 1855–1867.

[11] J.P. Robinson, M. Shao, Y. Wu, H. Wang, Y. Fu, Visual kinship recognition of families in the wild, *IEEE Trans. Pattern Anal. Mach. Intell.* 40 (11) (2018) 2624–2637.

[12] J.P. Robinson, M. Shao, Y. Fu, Recognizing families in the wild (RFIW): The 4th edition, in: Proc. CVPRW, 2020, pp. 858–866.

[13] I. Kemelmacher-Shlizerman, S.M. Seitz, D. Miller, E. Brossard, The MegaFace benchmark: 1 million faces for recognition at scale, in: Proc. CVPR, 2016, pp. 4873–4882.

[14] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, F.A. Wichmann, Shortcut learning in deep neural networks, *Nature Machine Intelligence* 2 (11) (2020) 665–673.